USAMO 2026 Solution Notes

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The Second Edition

Contents

55^{th} United States of America Mathematical Olympiad

Day I 1:00pm – 5:30pm ET

March 21, 2026

USAMO 1. Let n be an integer greater than 1. For which real numbers x is

$$\lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k}$$

maximal, and what is the maximal value that this expression can take?

Note: $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z.

USAMO 2. Annie is playing a game where she starts with a row of positive integers, written on a blackboard, each of which is a power of 2. On each turn, she can erase two adjacent numbers and replace them with a power of 2 that is greater than either of the erased numbers. This shortens the row of numbers, and she continues to take turns until only one number remains. Annie wins the game if the final remaining number is less than 4 times the sum of the original numbers. Is it always possible for Annie to win, regardless of the starting row of numbers?

USAMO 3. Let ABC be an acute scalene triangle with no angle equal to $60°$. Let ω be the circumcircle of ABC. Let $\triangle_B$ be the equilateral triangle with three vertices on ω, one of which is B. Let ℓ_B be the line through the two vertices of $\triangle_B$ other than B. Let $\triangle_C$ and ℓ_C be defined analogously. Let Y be the intersection of AC and ℓ_B, and let Z be the intersection of AB and ℓ_C.

Let N be the midpoint of minor arc BC on ω. Let $\mathscr{R}$ be the triangle formed by ℓ_B, ℓ_C, and the tangent to ω through N. Prove that the circumcircle of AYZ and the incircle of $\mathscr{R}$ are tangent.

55^{th} United States of America Mathematical Olympiad

Day II 1:00pm – 5:30pm ET

March 22, 2026

USAMO 4. A positive integer n is called *solitary* if, for any nonnegative integers a and b such that $a + b = n$, either a or b contains the digit 1 when written in base 10. Determine, with proof, the number of solitary integers less than 10^{2026}.

USAMO 5. Let ABC be a triangle. Points D, E, and F lie on sides BC, CA, and AB, respectively, such that

$$\angle AFE = \angle BDF = \angle CED.$$

Let O_A, O_B, and O_C be the circumcenters of triangles AFE, BDF, and CED, respectively. Let M, N, and O be the circumcenters of triangles ABC, DEF, and $O_AO_BO_C$, respectively. Prove that $OM = ON$.

USAMO 6. Let a and b be positive integers such that $\varphi(ab+1)$ divides $a^2 + b^2 + 1$. Prove that a and b are Fibonacci numbers.

Note: Euler's totient function $\varphi(n)$ is the number of positive integers less than or equal to n that are relatively prime to n.

1 Solutions to Day I

1.1 USAMO 2026 / Problem 1 Solution

Problem 1.1.1:

Let n be an integer greater than 1. For which real numbers x is

$$\lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k}$$

maximal, and what is the maximal value that this expression can take?

Note: $\lfloor z \rfloor$ denotes the greatest integer less than or equal to z.

★ *Proposed by Titu Andreescu and Gabriel Dospinescu*

1.1.1 Solution 1

The answer is $-1 + \sum_{k=1}^{n} \frac{1}{k}$. Let $f(x)$ denote the assertion.

Claim

It suffices to solve when $\lfloor x \rfloor = 0$.

Proof 1.1.1:

It suffices to show that the shift $x \to x - 1$ preserves the value. Indeed, we have

$$\begin{aligned}\lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k} \to \lfloor n(x-1) \rfloor - \sum_{k=1}^{n} \frac{\lfloor k(x-1) \rfloor}{k} \\ = \lfloor nx \rfloor - n - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor - k}{k} \\ = \lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k}\end{aligned}$$

so the value is preserved and our claim is proven. Thus, WLOG that $0 \leqslant x < 1$. $\square$

Claim

The maximum is achieved at $1 - \frac{1}{n} \leqslant x < 1$.

Proof 1.1.2:

If $x = 1 - \frac{1}{m}$ for a real $m \geqslant n$, then $\lfloor kx \rfloor = \lfloor k - \frac{k}{m} \rfloor = k - 1$ since $0 < \frac{k}{m} \leqslant 1$. $\square$

Claim

For each $0 \leqslant \ell < n$, we have that

$$\max_{\frac{\ell}{n} \leq x < \frac{\ell+1}{n}} f(x) = f\left(\frac{\ell}{n}\right).$$

Proof 1.1.3:

Note that as we move x from $\frac{\ell}{n} \to \frac{\ell+1}{n}$, we have that $\lfloor nx \rfloor$ is constant while $-\frac{\lfloor kx \rfloor}{k}$ can only decrease.

Thus, WLOG $x = \frac{\ell}{n}$ for an integer $0 \leqslant \ell < n$. Now, let $\{\bullet\} = \bullet - \lfloor \bullet \rfloor$. We rewrite as follows:

$$\begin{aligned}
\lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k} &= \ell - \sum_{k=1}^{n} \frac{\left\lfloor \frac{k\ell}{n} \right\rfloor}{k} \\
&= \sum_{k=1}^{n} \frac{\ell}{n} - \frac{\left\lfloor \frac{k\ell}{n} \right\rfloor}{k} \\
&= \sum_{k=1}^{n} \frac{\frac{k\ell}{n}}{k} - \frac{\left\lfloor \frac{k\ell}{n} \right\rfloor}{k} \\
&= \sum_{k=1}^{n} \frac{\left\{ \frac{k\ell}{n} \right\}}{k}
\end{aligned}$$

Let $g = \gcd(\ell, n)$. Note that

$$n\left\{\frac{k\ell}{n}\right\} \equiv k\ell \pmod{n}.$$

Furthermore,

$$g(k) = n\left\{\frac{k\ell}{n}\right\}$$

permutes $\left\{0, g, 2g, \ldots, \left(\frac{n}{g} - 1\right) g\right\}$ due to the above (2).
Now, write the sum as follows:

$$\sum_{k=1}^{n} \frac{\left\{\frac{k\ell}{n}\right\}}{k} = \sum_{k=1}^{\frac{n}{g}} \frac{\left\{\frac{k\ell}{n}\right\}}{k + \frac{mn}{g}} \text{ by (1)}$$

Now, due to (2) and rearrangement, we must have that

$$\sum_{k=1}^{\frac{n}{g}} \frac{\left\{\frac{k\ell}{n}\right\}}{k + \frac{mn}{g}} \leqslant \sum_{k=1}^{\frac{n}{g}} \frac{1 - \frac{kg}{n}}{k + \frac{mn}{g}}.$$

Thus, it suffices to show that

$$\sum_{m=0}^{g-1} \sum_{k=1}^{\frac{n}{g}} \frac{1 - \frac{kg}{n}}{k + \frac{mn}{g}} \leqslant \sum_{k=1}^{n} \frac{1}{k} - \frac{1}{n} = \sum_{m=0}^{g-1} \sum_{k=1}^{\frac{n}{g}} \frac{1}{k + \frac{mn}{g}} - \frac{1}{n}.$$

We sum by the columns (m), so it suffices to show that

$$\sum_{m=0}^{g-1} \frac{1 - \frac{kg}{n}}{k + \frac{mn}{g}} \leqslant \sum_{m=0}^{g-1} \frac{1}{k + \frac{mn}{g}} - \frac{1}{n}.$$

Indeed,

$$\sum_{m=0}^{g-1} \frac{1}{k + \frac{mn}{g}} - \frac{1}{n} \geqslant \sum_{m=1}^{g-1} \frac{1 - \frac{kg}{n}}{k + \frac{mn}{g}} + \frac{1}{k} - \frac{g}{n}$$

where we separated based on $m = 0$ and then we multiplied the other terms by $1 - \dfrac{kg}{n} < 1$. However, observe

$$\sum_{m=1}^{g-1} \frac{1 - \frac{kg}{n}}{k + \frac{mn}{g}} + \frac{1}{k} - \frac{g}{n} = \sum_{m=0}^{g-1} \frac{1 - \frac{kg}{n}}{k + \frac{mn}{g}}$$

and we are done.

Finally, we need to check the equality cases. Note that for the equality above, we need $m = 0$, so $g = 1$. Finally, for the rearrangement, we have

$$\left\{\frac{k\ell}{n}\right\} = 1 - \frac{k}{n}$$

which obviously occurs when $\ell = n - 1$. We are done. $\square$

1.1.2 Solution 2

We claim that the maximum value of the expression is $H_n - 1$, where H_n is the n-th harmonic number. The choices of x which produce this maximum are only the real numbers with fractional parts in the interval $\left[\frac{n-1}{n}, 1\right)$. First we show that these x truly achieve the maximum. Consider the following simplification:

Let $x = a + r$, where $a = \lfloor x \rfloor$ and $0 \le r < 1$. We have that

$$\frac{\lfloor kx \rfloor}{k} = \frac{ak + \lfloor kr \rfloor}{k} = a + \frac{\lfloor kr \rfloor}{k}.$$

Furthermore, $\lfloor nx \rfloor = an + \lfloor nr \rfloor$. Thus,

$$\lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k} = \lfloor nr \rfloor - \sum_{k=1}^{n} \frac{\lfloor kr \rfloor}{k}.$$

Now use that $r \in [\frac{n-1}{n}, 1)$:

- $\lfloor nr \rfloor = n - 1$.
- $\lfloor kr \rfloor = k - 1$.

Hence,

$$\lfloor nr \rfloor - \sum_{k=1}^{n} \frac{\lfloor kr \rfloor}{k} = n - 1 - \sum_{k=1}^{n} \frac{k-1}{k}$$

$$= n - 1 - \sum_{k=1}^{n} \left(1 - \frac{1}{k}\right) = -1 + \sum_{k=1}^{n} \frac{1}{k} = H_n - 1.$$

Now, we show that (1) this is the maximum value of the expression, and (2) the described x are the only x that achieves this value.

Define a **jump point** to be any real number $x = \frac{j}{k}$ (in simplest form), where $k \in \{1, 2, \ldots, n\}$, and j is a positive integer less than k. Between jump points, the expression is constant because none of the floor functions change; therefore, the maximum is attained in a half-open interval starting and ending with a jump point.

- At a jump point $\frac{j}{k}$ where kn, the value of the expression only decreases by $\frac{1}{k}$.
- At a jump point $\frac{j}{n}$, the value of the expression changes by $1 - \frac{1}{n} = \frac{n-1}{n}$.

Hence, the maximum must occur at some jump point $\frac{j}{n}$. The expression is then

$$j - \sum_{k=1}^{n} \frac{\left\lfloor k\frac{j}{n} \right\rfloor}{k}.$$

Then $\left\lfloor k\frac{j}{n} \right\rfloor = k\frac{j}{n} - \left\{k\frac{j}{n}\right\}$. Thus, we have

$$j - \sum_{k=1}^{n} \frac{k\frac{j}{n} - \left\{k\frac{j}{n}\right\}}{k} = \sum_{k=1}^{n} \frac{\left\{k\frac{j}{n}\right\}}{k}.$$

Note that $\left\{k\frac{j}{n}\right\} = \frac{kj \pmod n}{n}$.

Claim

Let $g = \gcd(n, j)$. The set $\{kj \pmod n), 1 \leqslant k \leqslant n\}$ contains the elements $\{0, g, 2g, \ldots, n-g\}$, each appearing exactly g times.

Proof 1.1.4:

Suppose $n = ga$ and $j = gb$ for a and b coprime. Then

$$kj \pmod n) = kgb \pmod{ga} = g(bk \pmod a)).$$

Because a and b are coprime, $bk \pmod a$ for $1 \leqslant k \leqslant a$ contains all residues modulo a. Because there are g "cycles" ($n = ga$) of this, each residue modulo a appears exactly g times. The desired result directly follows. $\square$

Claim

The maximum value of the expression occurs when $g = 1$.

Proof 1.1.5:

By Claim 1, the multiset

$$\left\{\{k\frac{j}{n}\}, 1 \leqslant k \leqslant n\right\}$$

is a permutation of the set

$$\left\{\frac{g}{n}, 2\frac{g}{n} \ldots, \frac{(n-g)}{n}\right\},$$

where each element appears exactly g times.

Now, define $a_k = \dfrac{1}{k}$ and $b_k = \left\{k\dfrac{j}{n}\right\}$. By the rearrangement inequality, the value of the expression $\displaystyle\sum_{k=1}^{n} a_k b_k$ (the desired sum) is maximized when a_k and b_k are similarly ordered. Because a_k is strictly decreasing, we seek a non-increasing b_k.

It is easy to see that b_k assorted in non-increasing order is equivalent to

$$b_k = \frac{n - g\left\lceil\frac{k}{g}\right\rceil}{n}.$$

Indeed, for $k = 1, 2, \ldots g$, $b_k = \dfrac{n-g}{n}$, and for $k = g+1, g+2, \ldots, 2g$, $b_k = \dfrac{n-2g}{n}$, and so on.

Because $k \leqslant g\left\lceil\dfrac{k}{g}\right\rceil$, each b_k is maximized when $g = 1$. In particular,

$$\frac{n-1}{n} > \frac{n-g}{n}$$

for $g > 1$, so there is at least one strict inequality. Hence, we have established that the strict maximum occurs with $g = 1$. $\square$

By Claim 2, we have established j is coprime to n. As a result, the multiset

$$\left\{\{k\frac{j}{n}\}, 1 \leqslant k \leqslant n\right\} \text{ is equivalent to } \left\{0, \frac{1}{n}, \frac{2}{n}, \ldots, \frac{n-1}{n}\right\}.$$

Applying the same rearrangement logic, we require the fractional parts to appear in strict decreasing order.

We claim that this is achieved solely at $j = n - 1$. This is easy to see because we would

require
$$\{1 \cdot \frac{j}{n}\} = \frac{n-1}{n} \iff j \equiv n-1 \pmod{n},$$
which, by the restrictions of j, force $j = n-1$.

As shown above, because $x = \dfrac{n-1}{n}$ in the original expression evaluates to $H_n - 1$, the maximum must be $H_n - 1$, achieved at all real numbers x whose fractional part is in the interval $\left[\dfrac{n-1}{n}, 1\right)$.

1.1.3 Solution 3

Let the given function be defined as:

$$f(x) = \lfloor nx \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k}$$

First, we show that $f(x)$ is periodic with period 1. Let $x = y + 1$. Then:

$$\begin{aligned}
f(x+1) &= \lfloor n(x+1) \rfloor - \sum_{k=1}^{n} \frac{\lfloor k(x+1) \rfloor}{k} \\
&= \lfloor nx + n \rfloor - \sum_{k=1}^{n} \frac{\lfloor kx + k \rfloor}{k} \\
&= \lfloor nx \rfloor + n - \sum_{k=1}^{n} \left(\frac{\lfloor kx \rfloor}{k} + 1 \right) \\
&= \lfloor nx \rfloor + n - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k} - n \\
&= f(x)
\end{aligned}$$

Because of this periodicity, we can restrict our analysis to $x \in [0, 1)$ and then generalize the result to all real numbers. We partition the interval $[0, 1)$ into n subintervals of the form $I_m = \left[\frac{m}{n}, \frac{m+1}{n} \right)$ for $m = 0, 1, \ldots, n-1$.

Let $x \in I_m$. This implies $m \leqslant nx < m + 1$, so $\lfloor nx \rfloor = m$. Additionally, since $x \geqslant \frac{m}{n}$ and $k > 0$, we have $kx \geqslant \frac{km}{n}$, which means $\lfloor kx \rfloor \geqslant \left\lfloor \frac{km}{n} \right\rfloor$ for all $k \in \{1, \ldots, n\}$.

Since $\lfloor nx \rfloor$ remains constant at m throughout I_m and the terms $\lfloor kx \rfloor$ are being subtracted, the maximum of $f(x)$ on I_m is bounded above by its value at the lower endpoint:

$$f(x) = m - \sum_{k=1}^{n} \frac{\lfloor kx \rfloor}{k} \leqslant m - \sum_{k=1}^{n} \frac{\left\lfloor \frac{km}{n} \right\rfloor}{k} = f\left(\frac{m}{n} \right)$$

This proves that the global maximum of $f(x)$ on $[0, 1)$ must be attained at one of the points $x = \frac{m}{n}$.

We now evaluate $f(\dfrac{m}{n})$. Using the identity $\lfloor z \rfloor = z - \{z\}$ (where $\{z\}$ is the fractional part of z), we get:

$$f\left(\frac{m}{n}\right) = m - \sum_{k=1}^{n} \frac{\frac{km}{n} - \left\{\frac{km}{n}\right\}}{k} = m - \sum_{k=1}^{n} \frac{m}{n} + \sum_{k=1}^{n} \frac{\left\{\frac{km}{n}\right\}}{k}$$

Since $\displaystyle\sum_{k=1}^{n} \frac{m}{n} = m$ and $\left\{\dfrac{nm}{n}\right\} = 0$, this simplifies beautifully to:

$$f\left(\frac{m}{n}\right) = \sum_{k=1}^{n-1} \frac{\left\{\frac{km}{n}\right\}}{k} = \frac{1}{n}\sum_{k=1}^{n-1} \frac{km \bmod n}{k}$$

Let $c_k = km \bmod n$. We want to find the integer $m \in \{0, \ldots, n-1\}$ that maximizes the sum $\displaystyle\sum_{k=1}^{n-1} \frac{c_k}{k}$.

We claim that the sum is strictly maximized when $m = n-1$. For $m = n-1$, the sequence becomes $c_k = k(n-1) \bmod n = -k \bmod n = n-k$.

To prove that any other m yields a smaller sum, we compare the sequence c_k with the sequence $b_k = n - k$. Let C be the multiset $\{c_1, \ldots, c_{n-1}\}$ and B be the set $\{1, 2, \ldots, n-1\}$.

Let $d = \gcd(m, n)$. The multiset C consists of the multiples of d from 0 to $n-d$. Specifically, 0 appears $d-1$ times, and each multiple vd (for $v \geqslant 1$) appears d times.

For any integer $y \in \{1, \ldots, n-1\}$, let $N_C(y)$ be the number of elements in C that are $\geqslant y$. The elements $vd \geqslant y \implies v \geqslant \left\lceil \dfrac{y}{d} \right\rceil$. The number of such valid multipliers v up to $\dfrac{n}{d} - 1$ is exactly $\dfrac{n}{d} - \left\lceil \dfrac{y}{d} \right\rceil$. Since each non-zero multiple appears d times, we have:

$$N_C(y) = d\left(\frac{n}{d} - \left\lceil \frac{y}{d} \right\rceil\right) = n - d\left\lceil \frac{y}{d} \right\rceil$$

Because $d\left\lceil \dfrac{y}{d} \right\rceil \geqslant d(\dfrac{y}{d}) = y$, we obtain $N_C(y) \leqslant n - y$. The number of elements in B that are $\geqslant y$ is precisely $N_B(y) = n - y$. Thus, $N_C(y) \leqslant N_B(y)$ for all y.

This establishes that the i-th largest element of C is $\leqslant$ the i-th largest element of B (which is $n-i$). Consequently, for any $j \in \{1, \ldots, n-1\}$, the sum of the first j elements of the sequence c_k is bounded by the sum of the j largest elements of C, giving:

$$\sum_{k=1}^{j} c_k \leqslant \sum_{k=1}^{j} (n-k)$$

Let $A_j = \sum_{k=1}^{j} (n-k-c_k)$. We have proven that $A_j \geqslant 0$ for all j. Using summation by parts on our sum:

$$\sum_{k=1}^{n-1} \frac{n-k-c_k}{k} = \sum_{j=1}^{n-2} A_j \left(\frac{1}{j} - \frac{1}{j+1}\right) + A_{n-1} \frac{1}{n-1}$$

Since $\frac{1}{j} - \frac{1}{j+1} > 0$ and $A_j \geqslant 0$, the entire sum is $\geqslant 0$. This implies $\sum_{k=1}^{n-1} \frac{c_k}{k} \leqslant \sum_{k=1}^{n-1} \frac{n-k}{k}$. Equality holds if and only if $A_j = 0$ for all j, meaning $c_k = n-k$ for all k, which occurs only when $c_1 = m = n-1$.

Since $m = n-1$ is the unique maximizer among our interval lower bounds, the maximum value of $f(x)$ evaluates to:

$$f\left(\frac{n-1}{n}\right) = \frac{1}{n} \sum_{k=1}^{n-1} \frac{n-k}{k} = \sum_{k=1}^{n-1} \frac{1}{k} - \frac{n-1}{n} = \sum_{k=1}^{n} \frac{1}{k} - 1$$

To find the continuous values of x where this maximum is achieved, we inspect the interval $I_{n-1} = \left[\frac{n-1}{n}, 1\right)$. For any x in this interval, and for $1 \leqslant k \leqslant n$:

$$k - \frac{k}{n} = k\left(\frac{n-1}{n}\right) \leqslant kx < k$$

Because $1 \leqslant k \leqslant n$, the term $\frac{k}{n} \leqslant 1$, meaning $k-1 \leqslant k - \frac{k}{n}$. Therefore, $kx \in [k-1, k)$, which locks $\lfloor kx \rfloor$ firmly at $k-1$. Computing $f(x)$ anywhere in this interval yields the constant maximum:

$$f(x) = (n-1) - \sum_{k=1}^{n} \frac{k-1}{k} = (n-1) - \sum_{k=1}^{n} \left(1 - \frac{1}{k}\right) = \sum_{k=1}^{n} \frac{1}{k} - 1$$

✓ **Conclusion**

Accounting for the periodicity of 1 established in Step 1, the maximal value is sustained across all intervals shifted by integers.

✓ **Maximal value:**

$$\sum_{k=1}^{n} \frac{1}{k} - 1$$

✓ **Real numbers x that achieve this:**

$$x \in \bigcup_{p \in \mathbb{Z}} \left[p - \frac{1}{n}, p\right)$$

1.2 USAMO 2026 / Problem 2 Solution

Problem 1.2.1:

Annie is playing a game where she starts with a row of positive integers, written on a blackboard, each of which is a power of 2. On each turn, she can erase two adjacent numbers and replace them with a power of 2 that is greater than either of the erased numbers. This shortens the row of numbers, and she continues to take turns until only one number remains. Annie wins the game if the final remaining number is less than 4 times the sum of the original numbers. Is it always possible for Annie to win, regardless of the starting row of numbers?

★ *Proposed by Andrew Dittmer*

1.2.1 Solution 1

We start by realizing that the algorithm of combining the smallest term with its smallest neighbor seems to work, and we attempt to prove that this algorithm yields a sum which is $\leqslant 4 \cdot$ (the original sum).

We proceed with strong induction with the base cases of $n = 1$ and $n = 2$ (if $n = 1$ our result is trivial and if $n = 2$ if the two numbers are x_1 and x_2, the combined term will be $2\max(x_1, x_2) \leqslant 2(x_1 + x_2)$).

Denote the original n integers as $x_1, x_2, \ldots, x_n$, and let

$$S = 2x_1 + 4x_2 + 4x_3 + 4x_4 + \cdots + 4x_{n-2} + 4x_{n-1} + 2x_n.$$

We attempt to prove that our result is always $\leqslant k$.

Let k be the first index such that

$$2x_1 + 4x_2 + \cdots + 4x_{k-1} + 4x_k = S_k \geqslant \frac{S}{2},$$

and assume that $k \neq 1$ or n.

WLOG, we can assume that $S_k - 2x_k \leqslant \frac{S}{2}$, as if not we can just mirror the sequence as both $S_k - 2x_k$ and $S - S_{k-1} - 2x_k$ cannot be greater than $\frac{S}{2}$, which contradicts our initial definition for S.

By our inductive hypothesis, as

$$S_k - 2x_k = 2x_1 + 4x_2 + \cdots + 4x_{k-1} + 2x_k,$$

we know we can combine the integers x_1 to x_k to form an integer $\leqslant \frac{S}{2}$, and similarly, as we know that the terms from x_{k+1} to x_n can be expressed an integer

$$\leqslant 2x_{k+1} + 4x_{k+2} + \cdots + 4x_{n-1} + 2x_n \leqslant S - S_k \leqslant \frac{S}{2},$$

implying that the combination of both terms $\leqslant S$, as desired.

In the case that $k = n$ (or if $k = 1$ we just mirror the entire sequence), we consider the sum of the first $n-1$ terms, and as we know $S_{n-1} - 2x_{n-1} \leqslant \frac{S}{2}$, we can combine the first $n-1$ terms to a value which is $\leqslant \frac{S}{2}$.

As we know $x_n \leqslant \frac{S}{2}$ (as $S \geqslant 2x_n$) we know that we can combine both terms to form a value $\leqslant S$, as desired. $\square$

1.2.2 Solution 2

Suppose the numbers on the blackboard are $2^{x_1}, 2^{x_2}, \ldots 2^{x_n}$.

Claim

Annie can always make the final number at maximum $-2 \cdot 2^{x_1} - 2 \cdot 2^{x_n} + 4\sum_{k=1}^{n} 2^{x_k}$.

Proof 1.2.1:

We proceed via strong induction on n, our trivial base case being when $n = 1$. Without loss of generality, suppose that x_v denotes the unique value such that

$$-2 \cdot 2^{x_1} - 2 \cdot 2^{x_v} + 4\sum_{k=1}^{v} 2^{x_k} \leqslant -2^{x_1} - 2^{x_n} + 2\sum_{k=1}^{n} 2^{x_k} \leqslant -2 \cdot 2^{x_1} + 4\sum_{v=1}^{n} 2^{x_v}.$$

As such, it follows that

$$-2 \cdot 2^{x_{v+1}} - 2 \cdot 2^{x_n} + 4\sum_{k=v+1}^{n} 2^{x_k} \leqslant -2^{x_1} - 2^{x_n} + 2\sum_{k=1}^{n} 2^{x_k}.$$

This means that combining the numbers $2^{x_1}, 2^{x_2}, \ldots 2^{x_v}$, we attain a number $\leqslant -2^{x_1} - 2^{x_n} + 2\sum_{k=1}^{n} 2^{x_k}$, and combining the numbers $2^{x_{v+1}}, 2^{x_{v+2}}, \ldots 2^{n}$, we attain a number $\leqslant -2^{x_1} - 2^{x_n} + 2\sum_{k=1}^{n} 2^{x_k}$, and combing these numbers results in the final number on our board being $\leqslant -2 \cdot 2^{x_1} - 2 \cdot 2^{x_n} + 4\sum_{k=1}^{n} 2^{x_k}$, as desired. $\square$

Remark

A strategy Annie can employ would be to first eliminate all pairs of adjacent 1s, then when moving onto the 2s eliminate all pairs of adjacent 2s and any remaining pairs of adjacent 2s and 1s, then when moving onto the 4s eliminate all pairs of adjacent 4s and any remaining pairs of adjacent 4s and 1s or 4s and 2s. Generalizing this to 2^j, when moving onto 2^j we eliminate all pairs of adjacent 2^js and any remaining pairs of adjacent 2^js and 1, 2^js and 2s, $\ldots$ any remaining pairs of adjacent 2^js and 2^{j-1}s.

1.2.3 Solution 3

Let $X = (x_1, x_2, \ldots, x_n)$ where $\forall i, x_i \in \{2^k \mid k \in \mathbb{Z}\}$, and let $S = \sum_{i=1}^{n} x_i$.

Define the adjacent merge operation $\omega(a, b) = 2\max(a, b)$.
Sequential adjacent applications of ω on $X \iff$ Construction of a full alphabetic (left-to-right) binary tree T.

- $\forall$ leaf $i \in T$: value $= x_i$, depth $= d_i$.
- $\forall$ internal node with children u, v: value $= \omega(u, v) = \max(2u, 2v)$.

By induction, the root value R is given by:

$$R = \max_{1 \leq i \leq n} (x_i 2^{d_i})$$

Winning condition $W \iff \exists$ valid tree T such that $R < 4S$.

$\because \forall i, x_i \in \{2^k\} \wedge$ closure of $\omega \implies R \in \{2^k \mid k \in \mathbb{Z}\}$.
Let $M = \max\{2^k \mid 2^k < 4S, k \in \mathbb{Z}\}$.

$$\begin{aligned}
\because R \in \{2^k\} \implies W &\iff R \leqslant M \\
&\iff \forall i,\ x_i 2^{d_i} \leqslant M \\
&\iff \forall i,\ 2^{d_i} \leqslant \frac{M}{x_i} \\
&\iff \forall i,\ d_i \leqslant \log_2 M - \log_2 x_i
\end{aligned}$$

Let $L_i = \log_2 M - \log_2 x_i$. ($\because M, x_i \in \{2^k\} \implies L_i \in \mathbb{Z}$).
$\implies W \iff \exists$ alphabetic binary tree T where $\forall i, d_i \leqslant L_i$.

Evaluate the Kraft sum constraint for $\{L_i\}$:

$$\sum_{i=1}^{n} 2^{-L_i} = \sum_{i=1}^{n} 2^{-\log_2(M/x_i)} = \sum_{i=1}^{n} \frac{x_i}{M} = \frac{S}{M}$$

By definition of M, we have $M < 4S \leqslant 2M$.

$$\implies \frac{1}{4} < \frac{S}{M} \leqslant \frac{1}{2} \implies \sum_{i=1}^{n} 2^{-L_i} \leqslant \frac{1}{2}$$

Let $q_i = 2^{-L_i+1} = 2 \cdot 2^{-L_i} \implies \sum_{i=1}^{n} q_i = 2 \sum_{i=1}^{n} 2^{-L_i} \leqslant 1$.

Define $C_1 = 0, C_{i+1} = C_i + q_i$, and intervals $I_i = [C_i, C_{i+1})$.

$\because \sum q_i \leqslant 1 \implies \forall i, I_i \subseteq [0,1] \wedge (i \neq j \implies I_i \cap I_j = \emptyset)$.

Notice $|I_i| = 2 \cdot 2^{-L_i}$. By the fundamental property of dyadic intervals:

$$\forall I \text{ with } |I| = 2 \cdot 2^{-L}, \exists \text{ sub-interval } J = [k \cdot 2^{-L}, (k+1) \cdot 2^{-L}) \subseteq I,\ k \in \mathbb{Z}$$

Let $k_i = \lceil C_i \cdot 2^{L_i} \rceil$, and define $J_i = [k_i \cdot 2^{-L_i}, (k_i + 1) \cdot 2^{-L_i})$.

$$\begin{aligned}
&\implies J_i \subseteq I_i \\
&\because I_i \cap I_j = \emptyset \wedge \text{strictly ordered left-to-right} \implies J_i \cap J_j = \emptyset \wedge \text{strictly ordered} \\
&\because J_i = [k_i 2^{-L_i}, (k_i+1) 2^{-L_i}) \iff \text{Unique binary prefix string of length } L_i \\
&\implies \{J_i\}_{i=1}^{n} \text{ forms a valid alphabetic prefix code} \\
&\implies \exists \text{ binary tree } T' \text{ with initial nodes } x_i \text{ at strict depth } L_i
\end{aligned}$$

Let $T = \text{Prune}(T')$ (recursively eliminate degree-1 internal nodes, redirecting child to parent).

$\implies T$ is a full alphabetic binary tree.

$\because$ Pruning strictly decreases or maintains depth without violating left-to-right order.

$$\begin{aligned}
&\implies \forall i \in T,\ \text{final depth } d_i \leqslant L_i \\
&\implies R = \max_{i=1}^{n}(x_i 2^{d_i}) \leqslant \max_{i=1}^{n}(x_i 2^{L_i}) = M < 4S \\
&\implies W \text{ is mathematically guaranteed.} \quad \square
\end{aligned}$$

1.3 USAMO 2026 / Problem 3 Solution

Problem 1.3.1:

Let ABC be an acute scalene triangle with no angle equal to $60°$. Let ω be the circumcircle of ABC. Let $\triangle_B$ be the equilateral triangle with three vertices on ω, one of which is B. Let ℓ_B be the line through the two vertices of $\triangle_B$ other than B. Let $\triangle_C$ and ℓ_C be defined analogously. Let Y be the intersection of AC and ℓ_B, and let Z be the intersection of AB and ℓ_C.

Let N be the midpoint of minor arc BC on ω. Let $\mathscr{R}$ be the triangle formed by ℓ_B, ℓ_C, and the tangent to ω through N. Prove that the circumcircle of AYZ and the incircle of $\mathscr{R}$ are tangent.

★ *Proposed by Rogelio Guerrero*

1.3.1 Solution 1

We use complex coordinates and let (ABC) be the unit circle. Write a, b, c, n for the coordinates of A, B, C, N. We have

$$|a| = |b| = |c| = |n| = 1 \quad \text{and} \quad n^2 = bc.$$

If we rotate b to 1, then the equation of ℓ_B is $\operatorname{Re}(z) = -1/2$. In general, the equation of ℓ_B is

$$\operatorname{Re}(z/b) = -1/2 \Longleftrightarrow \operatorname{Re}(\bar{b}z) = -1/2.$$

Similarly the equation of ℓ_C is

$$\operatorname{Re}(\bar{c}z) = -1/2.$$

The reflection across the perpendicular bisector of BC preserves the tangent at N and switches ℓ_B and ℓ_C. Thus $\mathscr{R}$ is an isosceles triangle and its incenter I lies on ON where O is the origin.

Write λn as the coordinate of the incenter where λ is a real number. We have $\lambda < 1$ and the inradius of $\mathscr{R}$ is $1 - \lambda$. To compute the distance from I to ℓ_B we rotate b to 1 again. The rotation of I is $\lambda n/b$ and the distance to the rotation of ℓ_B (which is $\operatorname{Re}(z) = -1/2$) is

$\mathrm{Re}(\lambda n/b) + 1/2$. Since $\mathrm{Re}(n/b) = \cos A$ (the arc from B to N has central angle A),

$$\lambda \cos A + 1/2 = 1 - \lambda \implies \lambda = \frac{1}{2(1+\cos A)}.$$

The incircle has equation

$$|z - \lambda n| = 1 - \lambda.$$

To prove the tangency, we invert with center A and power 1:

$$z \mapsto z^* \quad \text{such that} \quad z^* - a = \frac{1}{\bar{z} - \bar{a}}.$$

The circle (AYZ) maps to a line Y^*Z^* and the incircle of $\mathscr{R}$ maps to a circle ω'. It suffices to show that Y^*Z^* is tangent to ω'.

A line through two points p, q on the unit circle has equation $z + pq\bar{z} = p + q$. So Y satisfies

$$z + ac\bar{z} = a + c,$$
$$z + b^2\bar{z} = -b.$$

Subtracting gives $\bar{Y} = s/(ac - b^2)$ where $s = a + b + c$. Thus

$$Y^* = \frac{a^2 s}{a^2 + ab + b^2}, \quad Z^* = \frac{a^2 s}{a^2 + ac + c^2}.$$

Lemma 1.3.1:

Let $|z - p| = r$ be a circle centered at p with $|p| \neq r$. Its inversion across the unit circle has center $p' = p/\mathrm{pow}$ and radius $r' = r/\mathrm{pow}$ where $\mathrm{pow} = |p|^2 - r^2$.

Proof 1.3.1:

The inverted circle is

$$|1/\bar{z} - p| = r,$$

which rewrites as

$$\left| z - \frac{p}{|p|^2 - r^2} \right|^2 = \frac{r^2}{(|p|^2 - r^2)^2}.$$

$\square$

Using a coordinate change we have

Corollary 1.3.1:

If we invert a circle ω centered at z_0 across a unit circle centered at a, then the center of the inverted circle is

$$z_0' = a + \frac{z_0 - a}{\text{pow}},$$

where pow is the power of a with respect to ω.

For the incircle (center λn, radius $1-\lambda$), the power is

$$|\lambda n - a|^2 - (1-\lambda)^2 = \lambda |n-a|^2,$$

and the center is

$$\begin{aligned} W &= a + \frac{\lambda n - a}{\text{pow}} = a + \frac{\lambda n - a}{\lambda |n-a|^2} \\ &= a + \frac{an(\lambda n - a)}{-\lambda (n-a)^2} = \frac{a^2((1-2\lambda)n + \lambda a)}{\lambda (n-a)^2}. \end{aligned}$$

Since $(1-2\lambda)/\lambda = 2\cos A = (b+c)/n$, we have

$$W = \frac{a^2 s}{(n-a)^2}.$$

And the radius is $\rho = (1-\lambda)/(\lambda |n-a|^2)$.

Write $D_b = a^2 + ab + b^2, D_c = a^2 + ac + c^2$. Dividing W, Y^*, Z^* by $a^2 s$, it suffices to show that the distance from

$$P = \frac{1}{(n-a)^2}$$

to the line through $Q = 1/D_b$ and $R = 1/D_c$ equals

$$\frac{\rho}{|a^2 s|} = \frac{1 + 2\cos A}{|n-a|^2 |s|}.$$

The distance is $|\mathscr{D}|/(2|Q-R|)$ where

$$\mathscr{D} = \det \begin{pmatrix} P & \bar{P} & 1 \\ Q & \bar{Q} & 1 \\ R & \bar{R} & 1 \end{pmatrix}.$$

Since a, b, c, n lie on the unit circle,

$$\bar{P} = a^2 n^2 P, \quad \bar{Q} = a^2 b^2 Q, \quad \bar{R} = a^2 c^2 R.$$

Factoring from each row, then a^2 from column 2, we have:

$$\begin{aligned} \mathscr{D} &= \frac{a^2}{(n-a)^2 D_b D_c} \det \begin{pmatrix} 1 & n^2 & (n-a)^2 \\ 1 & b^2 & D_b \\ 1 & c^2 & D_c \end{pmatrix} \\ &= \frac{a^3}{(n-a)^2 D_b D_c} \det \begin{pmatrix} 1 & n^2 & -2n \\ 1 & b^2 & b \\ 1 & c^2 & c \end{pmatrix} \\ &= \frac{a^3(b-c)}{(n-a)^2 D_b D_c} \det \begin{pmatrix} n^2 - c^2 & -2n - c \\ b + c & 1 \end{pmatrix} \\ &= \frac{2a^3 n(b-c)(b+c+n)}{(n-a)^2 D_b D_c}. \end{aligned}$$

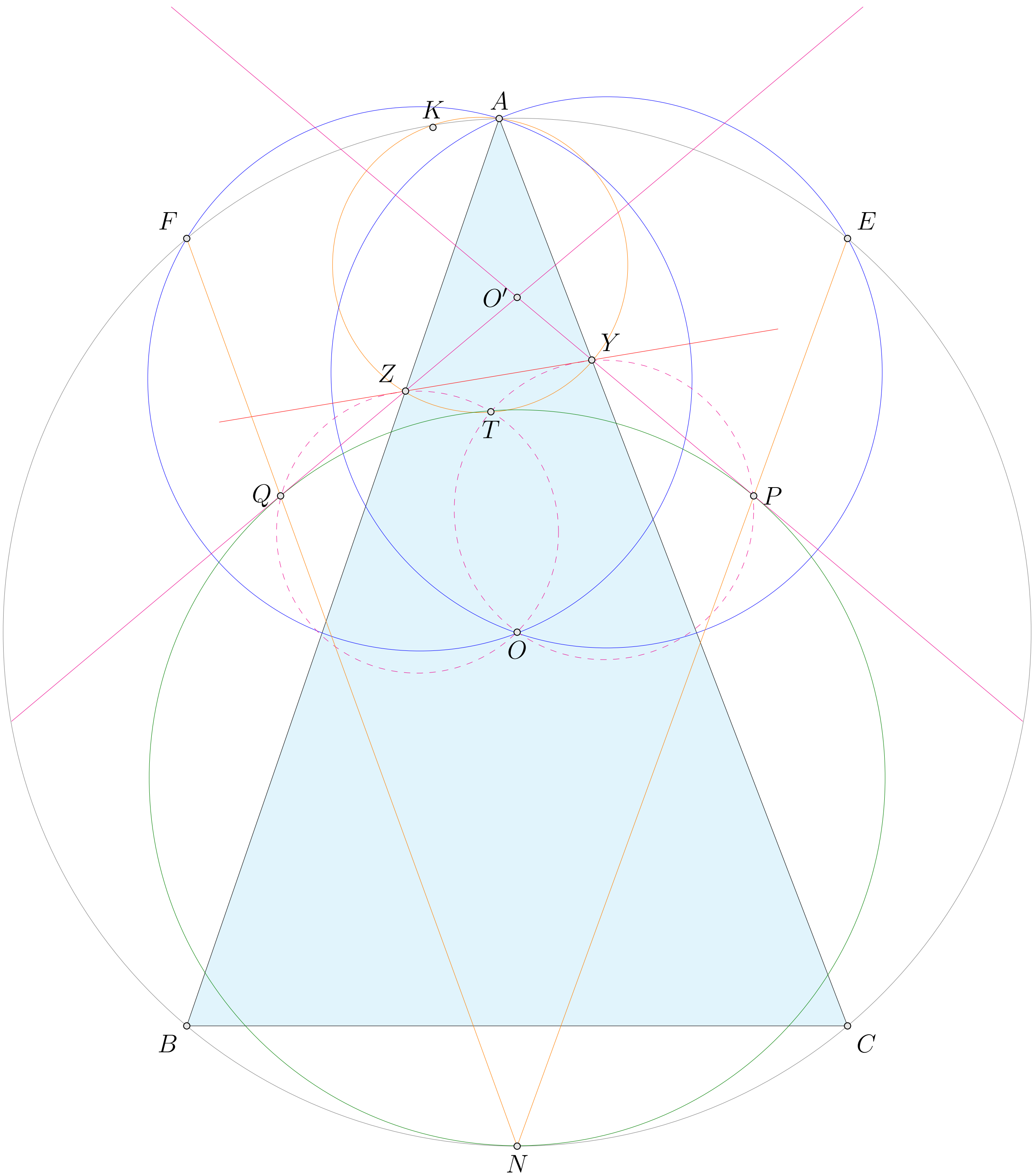
where the last step uses $n^2 = bc$.

Using $|Q-R| = |b-c||s|/(|D_b||D_c|)$ and $|a| = |n| = 1$:

$$\begin{aligned} \operatorname{dist}(P, QR) = \frac{|\mathscr{D}|}{2|Q-R|} &= \frac{|b+c+n|}{|n-a|^2|s|} \\ &= \frac{1 + 2\cos A}{|n-a|^2|s|} = \frac{\rho}{|a^2 s|}. \end{aligned}$$

Hence we win. $\square$

1.3.2 Solution 2



Let O be the circumcenter of $\triangle ABC$. Let E, F be the reflection of B, C across O respectively. Note that ℓ_B, ℓ_C are bisectors of OF, OE, respectively. Let O' be the circumcenter of $\triangle OEF$.

Claim

Let K be the other intersection of $\odot(AYZ)$ and $\odot(ABC)$. Then, Y, Z are circumcenters of $\triangle KOE$ and $\triangle KOF$, respectively.

Proof 1.3.2:

Let Y', Z' be circumcenters of $\triangle KOE$ and $\triangle KOF$, respectively.

Note that

$$2\measuredangle Y'KZ' = -\measuredangle EOF = -\measuredangle Y'O'Z'.$$

Moreover,

$$\measuredangle KYZ = \measuredangle KAZ = \measuredangle KEO = \measuredangle KY'Z'.$$

Similarly, $\measuredangle KZY = \measuredangle KZ'Y'$. Hence, $\triangle KYZ \overset{+}{\sim} \triangle KY'Z'$.

Suppose $YZ \neq Y'Z'$, we use spiral similarity, so Y', Z', K, O' are concyclic. Combined with the angle chasing above, we have $2\measuredangle Y'KZ' = -\measuredangle Y'KZ'$ That is, $3\measuredangle Y'KZ' = 0$, which will cause $\angle A = 60°$ or $\angle A = 120°$, which is a contradiction. Therefore, $Y = Y', Z = Z'$. $\square$

The claim suggests that YZ is the bisector of KO. Thus, O is the excenter of $\triangle O'YZ$. Let ω, the incircle of $\mathscr{R}$, touch ℓ_B, ℓ_C at P, Q.

Claim

Let $\angle PO'O = \alpha$, then $\angle O'PO = 45° + \alpha/2$.

Proof 1.3.3:

By homothety from ω to the circumcircle of $\triangle ABC$, we have N, P, E are collinear. Because $PO = PE$ and $OE = OF$, we have

$$\angle O'PO = 90° - \angle ONE = 90° - \angle O'OE/2 = 45° + \angle O'OP/2. \qquad \square$$

Let T be the other intersection of $\odot(OZQ)$ and $\odot(OYP)$.

Because

$$\angle YTP = \angle PYO + \angle YQO = 90^\circ + \alpha = 180^\circ - \angle YAZ.$$

We have T lies on $\odot(AYZ)$. Moreover, the previous claim suggests that

$$\begin{aligned}
\angle PTQ &= \angle PTO + \angle QTO \\
&= \angle PYO + \angle QYO \\
&= \angle YO'Z + \angle YOZ \\
&= 2\alpha + (90^\circ - \alpha) \\
&= 90^\circ + \alpha.
\end{aligned}$$

Since $\angle O'PQ = 90^\circ - \alpha$. By tangent line theorem, T lies on ω.

To finish this, angle chase to obtain that $\angle PTY = \angle PQT + \angle TZY$ and imply that $\odot(AYZ)$ and ω are tangent at T.

Remark

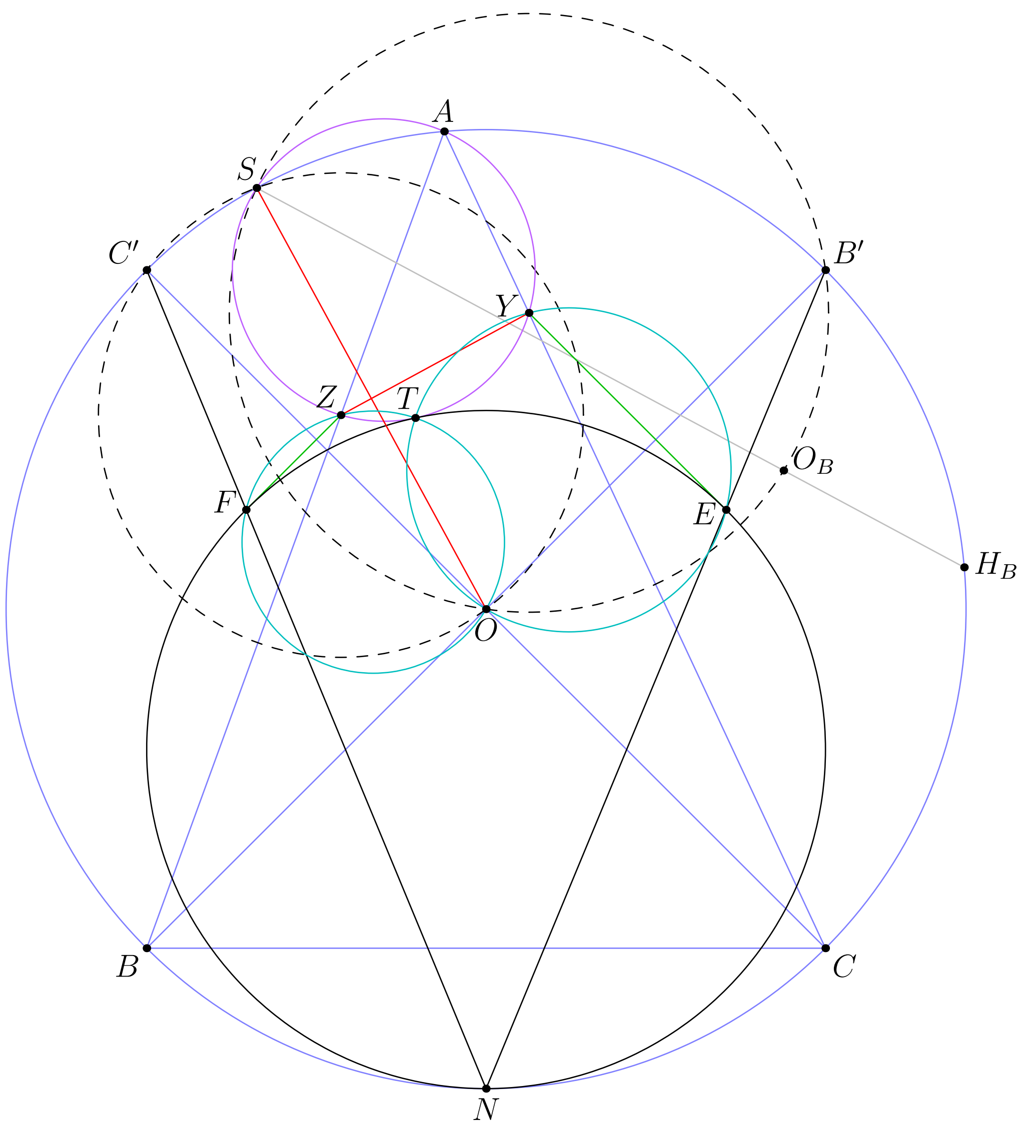
This problem as two main parts. First, to figuring out that YZ is the bisector of KO. Then, you can simply remove many points, considering O', Y, Z and ω and the incircle of $\mathscr{R}$ as main conditions. This configuration is very similar to the mixtillinear lemma. (That's how I come up that T the tangent point is the intersection of two circles.)

1.3.3 Solution 3

Let O be the circumcenter of (ABC). Let O_B and H_B be the reflections of O and the orthocenter over $\overline{AC}$. If B' is the antipode of B, then ℓ_B is the perpendicular bisector of $\overline{OB'}$. Then Y is the center of (OO_BB').

Let S be the second intersection of $\overline{H_BO_B}$ with (ABC). Note that $S \in (OO_BB')$ by Reim's on $\overline{SO_BH_B}$ and $\overline{B'OB}$ with $\overline{O_BO} \parallel \overline{H_BB}$. Also, by definition S is the anti-Steiner point of the Euler line. By symmetry, Z is the center of $(OC'S)$. This means S is the reflection of O over $\overline{YZ}$.

Let E and F be the tangency points of the incircle of $\mathscr{R}$ to ℓ_B and ℓ_C respectively. By the Shooting lemma, $\overline{NE}$ intersects (ABC) again at B' because the tangent to (ABC) at B' is parallel to ℓ_B.



We claim the desired tangency point is T, the second intersection of (OEY) and (OFZ). To show T lies on ω, angle chase

$$\begin{aligned}\measuredangle YTZ = \measuredangle YTO + \measuredangle OTZ &= \measuredangle YEO + \measuredangle OFZ \\ &= \measuredangle B'EY + \measuredangle ZFC' \\ &= (90^\circ - \measuredangle OB'E) + (90^\circ - \measuredangle FC'O) \\ &= \measuredangle NB'B + \measuredangle CC'N \\ &= \measuredangle CAB = \measuredangle YAZ.\end{aligned}$$

To show it lies on the incircle of $\mathscr{R}$, angle chase

$$\begin{aligned}\measuredangle ETF = \measuredangle ETO + \measuredangle OTF &= \measuredangle EYO + \measuredangle OZF \\ &= \measuredangle B'SO + \measuredangle OSC' \\ &= \measuredangle B'SC' = \measuredangle B'NC' = \measuredangle ENF.\end{aligned}$$

Finally, to show the circles are tangent at T it suffices to show

$$\measuredangle YTE + \measuredangle ZTF = \measuredangle(\overline{YZ}, \overline{EF}).$$

Angle chase again using $\overline{OY}$ is the perpendicular bisector of $\overline{B'S}$ to get

$$\begin{aligned}\measuredangle YTE &= \measuredangle YOE \\ &= \measuredangle YOB' + \measuredangle B'OE \\ &= \measuredangle SOY + \measuredangle EB'O \\ &= \measuredangle SOY + \measuredangle NAB.\end{aligned}$$

Therefore,

$$\begin{aligned}\measuredangle YTE + \measuredangle ZTF &= \measuredangle SOY + \measuredangle NAB + \measuredangle SOZ + \measuredangle NAC \\ &= (90^\circ - \measuredangle OB'S) + (90^\circ - \measuredangle OC'S) \\ &= \measuredangle SB'O + \measuredangle SC'O \\ &= \measuredangle SB'C' + \measuredangle SC'B'.\end{aligned}$$

Now, if ℓ_S is the tangent to (ABC) at S, then we have $\ell_S \parallel \overline{YZ}$ and $\overline{B'C'} \parallel \overline{EF}$. Finish with

$$\measuredangle SB'C' + \measuredangle SC'B' = \measuredangle(\ell_S, \overline{B'C'}) = \measuredangle(\overline{YZ}, \overline{EF}). \quad \square$$

Remark

Some more concurrencies:

- $\overline{B'C'}$, $\overline{OY}$, (AYZ), (OSC') and similarly for $\overline{OZ}$.
- $\overline{EF}$, $\overline{OY}$, and (OTZ) and similarly for $\overline{OZ}$.
- $\overline{NE}$, (OTY), (OSB') and similarly for $\overline{NF}$.
- The two points in the previous one are collinear with O. In fact, they lie on an angle bisector of $\angle YOZ$.

1.3.4 Solution 4

Line ℓ_B has equation $z + b^2\overline{z} = -b$, so $y = \dfrac{b(ab+bc+ca)}{b^2-ac}$ and

$$y - a = \frac{c(a^2+ab+b^2)}{b^2-ac}.$$

Therefore, the $\sqrt{bc}$ inverse of Y is

$$\frac{(b-a)(c-a)(b^2-ac)}{c(a^2+ab+b^2)} + a.$$

The difference of the $\sqrt{bc}$ inverses of Y and Z is

$$\frac{(a-b)(a-c)(b-c)(ab+bc+ca)^2}{bc(a^2+ab+b^2)(a^2+ac+c^2)}$$

so the line between them is

$$(a+b+c)^2 z + (ab+bc+ca)^2\overline{z} = \frac{a^2(b+c)^3 + a(-b^4+b^3c+2b^2c^2+bc^3-c^4) + bc(b+c)^3}{bc}.$$

This intersects line BC at

$$((ab+bc+ca)^2 - bc(a+b+c)^2)\overline{z} = \frac{a(b^2+bc+c^2)(a(b+c)-(b^2+c^2))}{bc}$$

so

$$\overline{z} = \frac{a(a(b+c)-(b^2+c^2))}{bc(a^2-bc)}.$$

This means

$$z = \frac{a(b^2+c^2)-bc(b+c)}{a^2-bc}.$$

Since the incircle is tangent to ω at N, we want to show that the center of the inverted incircle is on the angle bisector of BC and the inverted line YZ. Note that the angle bisector has equation $(a+b+c)z \pm \sqrt{bc}(ab+bc+ca)\overline{z}$.

The tangent to $-c$ has equation $z + c^2\overline{z} = -2c$, the line ℓ_C has equation $z + c^2\overline{z} = -c$, and the line parallel through N has equation $z + c^2\overline{z} = \dfrac{c(b+c)}{\sqrt{bc}}$. This means that the ratio of homothety is equal to

$$\frac{\frac{c(b+c)}{\sqrt{bc}} + c}{\frac{c(b+c)}{\sqrt{bc}} + 2c} = 1 - \frac{\sqrt{bc}}{b+c+2\sqrt{bc}}.$$

Therefore, the center of the circle is

$$\frac{bc}{b+c+2\sqrt{bc}}.$$

After $\sqrt{bc}$ inversion, the center lies on the line through A and $b+c+2\sqrt{bc}$, and the circle is tangent to BC at $AN \cap BC$. Consider the $z + bc\overline{z}$ coordinate of the three points. A has coordinate $\dfrac{a^2+bc}{a}$. The point on BC has coordinate $b+c$. N has coordinate $2\sqrt{bc}$. The ratio $\dfrac{A(AN \cap BC)}{AN}$ is $\dfrac{(b-a)(c-a)}{(a-\sqrt{bc})^2}$. This means that the center of the new circle is

$$(-a+b+c+2\sqrt{bc})\frac{(b-a)(c-a)}{(a-\sqrt{bc})^2}+a.$$

Subtract this with the point from earlier to get

$$\frac{2(a-b)(a-c)(ab+bc+ca)}{(a^2-bc)(a-\sqrt{bc})}.$$

The ratio of this point and its conjugate is

$$\frac{\sqrt{bc}(ab+bc+ca)}{a+b+c}$$

which is exactly what we wanted to show. $\square$

2 Solutions to Day II

2.1 USAMO 2026 / Problem 4 Solution

Problem 2.1.1:

A positive integer n is called *solitary* if, for any nonnegative integers a and b such that $a+b=n$, either a or b contains the digit 1 when written in base 10. Determine, with proof, the number of solitary integers less than 10^{2026}.

★ *Proposed by Carl Schildkraut*

2.1.1 Solution 1

Lemma 2.1.1:

$n = 2 \cdot 10^k - 1$ is solitary for all $k \in \mathbb{Z}_{\geqslant 0}$.

Proof 2.1.1:

Suppose $a+b=n$. WLOG, $a \geqslant b$. Then,

$$2a \geqslant a+b=n \implies a \geqslant \frac{n}{2}.$$

Since a, n are integers, and n is odd, this implies

$$a \geqslant \frac{n+1}{2} = 10^k.$$

But, $a \leqslant n = 2 \cdot 10^k - 1$, so $10^k \leqslant a \leqslant 2 \cdot 10^k - 1$, so the leading digit of a is 1. Thus, n is solitary. $\square$

Lemma 2.1.2:

$m = 2 \cdot 10^k + n, n < 10^k$ is solitary if and only if n is solitary.

Proof 2.1.2:

First, suppose m is solitary and n is not. We write $n = a + b$ for some a, b with no 1's in their decimal representation. Then, let $a' = a + 2 \cdot 10^k$ and $b' = b$. It follows that

$$a' + b' = a + 2 \cdot 10^k + b = 2 \cdot 10^k + n = m$$

But a' and b' have no 1's in their decimal representation, since $10^k > n \geqslant a, b$, a contradiction. This proves the only if part.

Now, suppose n is solitary, and m is not. Write $m = a + b$ for some a, b with no 1's in their decimal representation.

Write $a = x \cdot 10^k + a', a' < 10^k$ (the leading digit of a is x). Similarly, $b = y \cdot 10^k + b', b' < 10^k$. By definition, $a, b \neq 1$. But

$$x = y = 0 \implies a + b < 2 \cdot 10^k < n \text{ and } x + y > 2 \implies a + b \geqslant 3 \cdot 10^k > m$$

Thus, one of x, y is 2, and the other is 0. WLOG $x = 2, y = 0$. Consider a' and b'. We have

$$a' + b' = m - x \cdot 10^k - y \cdot 10^k = m - 2 \cdot 10^k = n$$

But n is solitary, so either a' or b' has a 1 in their decimal representation. But, that implies a or b also has a 1, since $a', b' < 10^k$, a contradiction. This proves the if part. The lemma is complete. □

Lemma 2.1.3:

For any integer $m \in [10^k, 10^{k+1} - 1]$, if m is not of the form $2 \cdot 10^k - 1$ and not of the form $2 \cdot 10^k + n$ (where $n < 10^k$), then m is not solitary.

Proof 2.1.3:

By Lemma 2.1.2, we can assume the leading digit of m is not 2. Now, take $a = 10^k - 1$ and $b = m - a$.

Let the decimal representations of a, b be $a_{k-1}a_{k-2}\ldots a_0$ and $b_k b_{k-1}\ldots b_0$, respectively. It follows that $a_i = 9$ for all i.

Also, $m \notin [2 \cdot 10^k - 1, 3 \cdot 10^k - 1] \implies b = m - a \notin [10^k, 2 \cdot 10^k]$, so $b_k \neq 1$.

Now, suppose $b_i = 1, i \in \{0, 1, \dots, k-1\}$. Increase b_i to 2 and lower a_i to 8. This increases b by 10^i while decreasing a by 10^i, so the sum stays the same. Repeating this for all i such that $b_i = 1$, we have a, b such that none of the digits are 1, $a+b = m$, so m is not solitary. $\square$

Now, we use the lemmas to finish. Let s_k be the number of solitary integers $< 10^k$, and let S_k be the set of these integers, so $|S_k| = s_k$. By the lemmas

$$S_{k+1} = \{2 \cdot 10^k - 1\} \cup \{2 \cdot 10^k + n : n \in S_k\} \cup S_k$$

Since no elements are shared, we have $s_{k+1} = 1 + s_k + s_k = 2s_k + 1$. Also, $S_1 = \{1\}$, so $s_1 = 1$. It follows by induction $s_k = 2^k - 1$. Thus, the answer is $\boxed{2^{2026} - 1}$.

2.1.2 Solution 2

We claim that a number is solitary if and only if all the following hold:

- The digit 1 appears exactly once;
- Every digit (possibly none) to the left of the 1 is 0 or 2;
- Every digit (possibly none) to the right of the 1 is 9.

For example, 202201999999 is solitary.

★ **Proof all such numbers are solitary**

The basic idea is to use induction as follows: If the last digit of n is 9, then the last digits of a and b sum to 9; hence we can ignore the last digit altogether. Thus, we reduce to the case where the last digit is 1. We continue the induction in a similar way in this situation. Zero-pad all the numbers to be the same length as n. Take the leading 2 of n.

- If either or a and b has a leading digit 2, we can delete it and continue the induction.
- Otherwise, clearly one of a or b must have leading digit 1, as needed.

★ **Proof that every solitary number is of this form**

Let n be solitary. In all the diagrams that follow, ellipses denote groups of digits other than 1 (possibly none). We first reduce to the case where n has exactly one 1:

- By taking $b = 0$, we see there is at least a single 1.
- Suppose n has an even number of 1's. The idea is to pair the 1's using blocks like $9 \ldots 9$, as in the following diagram:

$$\begin{aligned} n &= \quad \ldots 1 \quad \ldots 1 \quad \ldots 1 \quad \ldots 1 \quad \ldots \\ a &= \quad 0000 \quad 9993 \quad 0000 \quad 9993 \quad 0000 \\ b &= \quad \ldots 0 \quad \ldots 8 \quad 0000 \quad \ldots 8 \quad \ldots \end{aligned}$$

- Next, suppose n has an odd number of 1's and at least three 1's. The strategy is similar:

$$\begin{aligned} n &= \quad \ldots 1 \quad \ldots 1 \quad \ldots 1 \quad \ldots 1 \quad \ldots 1 \quad \ldots \\ a &= \quad 0000 \quad 9993 \quad 9993 \quad 0000 \quad 9993 \quad 0000 \\ b &= \quad \ldots 1 \quad \ldots 7 \quad \ldots 8 \quad 0000 \quad \ldots 8 \quad \ldots \end{aligned}$$

So, assume n has exactly one 1 appear.

- If a digit $b \neq 0, 9$ appears to the left of 1, let $c = b + 1$ to get

$$\begin{aligned} n &= \quad \ldots \quad 1 \quad \ldots b \quad \ldots \\ a &= 0000 \quad 0 \quad 9999 \quad 0000 \\ b &= \quad \ldots \quad 0 \quad \ldots c \quad \ldots \end{aligned}$$

  If $b = 0$, one can instead replace 9999 with 9998 and choose $c = 2$.

- Finally, if a digit $d \neq 0, 2$ appears to the left of 1, let $e = d - 1$ and use a bunch of 9's:

$$\begin{aligned} n &= \ldots d \quad \ldots \quad 1 \quad \ldots \\ a &= 0000 \quad 9999 \quad 9 \quad 0000 \\ a &= \ldots e \quad \ldots \quad 2 \quad \ldots \end{aligned}$$

★ Final Count

Zero-pad the number so that it has exactly 2026 digits. If we pick the digit 1 to appear at the i'th place from the left, for $1 \leqslant i \leqslant 2026$, then there are 2^{i-1} ways to pick the first $i - 1$ digits (either 0 or 2); the rest are all 9. Hence, the answer is

$$\sum_{i=1}^{2026} 2^{i-1} = \boxed{2^{2026} - 1}.$$

2.1.3 Solution 3

Let the following numbers be written as

$$\begin{aligned} a &= \overline{a_{d(a)-1}a_{d(a)-2}\ldots a_0} \\ b &= \overline{b_{d(b)-1}b_{d(b)-2}\ldots b_0} \\ n &= \overline{n_{d(n)-1}n_{d(n)-2}\ldots n_0} \end{aligned}$$

where $d(k)$ is the number of digits in k. We will aim to force n to be solitary by eliminating cases of a and b that disprove it. Let k be the smallest positive integer such that $n_k \neq 0$.

For $n_0 = 1$:

If $a_0+b_0 = 1$, then $(a_0, b_0) = (0, 1)$ or $(a_0, b_0) = (1, 0)$, which does not disprove. If $a_0+b_0 = 11$, then we can make $(a_j, b_j) = (2, 7)$ for all $1 \leqslant j < k$, forcing $\overline{n_{d(n)-1}n_{d(n)-2}\ldots(n_k-1)} = \left\lfloor \frac{n}{10^k} \right\rfloor - 1$ to be solitary. This is necessary and sufficient.

For $n_0 = 9$:

Note that $a_0 + b_0 = 9$ because a carryover is impossible here. If we let $(a_0, b_0) = (2, 7)$, then $\overline{n_{d(n)-1}n_{d(n)-2}\ldots n_1} = \left\lfloor \frac{n}{10} \right\rfloor$ must be solitary. Once again, this is necessary and sufficient.

For $n_0 = 0$ or $1 < n_0 < 9$:

If $a_0 + b_0 = n_0$, then we can make it so $a_0 \neq 1$ and $b_0 \neq 1$, forcing $\left\lfloor \frac{n}{10} \right\rfloor$ to be solitary.

If $a_0+b_0 = 10+n_0$, then we can let $(a_0, b_0) = (2, 7)$, so $\overline{n_{d(n)-1}n_{d(n)-2}\ldots(n_k-1)} = \left\lfloor \frac{n}{10^k} \right\rfloor - 1$ must be solitary. These are both necessary and sufficient.

Lemma 2.1.4:

If $n_0 = 0$ or $1 < n_0 < 9$, then $\left\lfloor \frac{n}{10^k} \right\rfloor$ must be *solitary*.

Proof 2.1.4:

Note that if $10s$ is solitary, so must s. This follows directly from above. The rest follows from induction. $\square$

Claim

If all *solitary* numbers less than 10^k have a unit digit of 1 or 9, then so will all solitary numbers less than 10^{k+1}.

Proof 2.1.5:

Note that the first *solitary* number is 1. It's necessary and sufficient for both $\left\lfloor \frac{n}{10^k} \right\rfloor$ and $\left\lfloor \frac{n}{10^k} \right\rfloor - 1$ to be solitary if a number with $k+1$ digits has a unit digit not 1 or 9. However, by our hypothesis, this cannot exist since numbers ending with digits 1 or 9 cannot be consecutive. $\square$

To finish this off, we do casework on the last digit.

If $d_0 = 1$:

The number $\left\lfloor \frac{n}{10^k} \right\rfloor - 1$ must be solitary with the last digit not equal to 9 (or else $d_k = 0$). However, any number ending with 1 suffices. Let $p(d)$ be the number of solitary numbers with d digits ending with 1. Then by doing casework on k, we have

$$p(d) = \sum_{k=1}^{d-1} p(k)$$
$$p(1) = 1$$

Solving this gives $p(d) = 2^{d-2}$.

If $d_0 = 9$:

The number $\left\lfloor \frac{n}{10} \right\rfloor$ must be solitary. Note that if $k > 1$, then for $\left\lfloor \frac{n}{10} \right\rfloor$ to be solitary, both $\left\lfloor \frac{n}{10^k} \right\rfloor$ and $\left\lfloor \frac{n}{10^k} \right\rfloor - 1$ must be solitary, but by Claim, we have already shown it is impossible. If $k = 1$, only $\left\lfloor \frac{n}{10} \right\rfloor$ needs to be solitary, so all solitary numbers with $d(n) - 1$ digits work. The difference is that there is no extra 0 that would lead to more restrictions. Let $q(d)$ be the number of solitary numbers with d digits ending with 9. Computing, we have

$$q(d) = p(d-1) + q(d-1) = 2^{k-3} + q(d-1)$$

And we have

$$q(1) = 0$$
$$q(2) = 1$$
$$q(3) = 2$$

Solving this gives $q(d) = 2^{k-2}$. We want

$$\begin{aligned}
\sum_{k=1}^{2026} [p(k) + q(k)] &= \sum_{k=1}^{2026} p(k) + q(1) + \sum_{k=2}^{2026} q(k) \\
&= p(2027) + 0 + \sum_{k=0}^{2024} 2^k \\
&= 2^{2025} + (2^{2025} - 1) \\
&= \boxed{2^{2026} - 1}
\end{aligned}$$

2.1.4 Solution 4

Generalize to all positive integers. I claim all solitary numbers are of the form $2 \cdot 10^{a_n} + 2 \cdot 10^{a_{n-1}} + \dots 2 \cdot 10^{a_1} \pm 1$ for some $1 \leqslant a_1 < a_2 < \dots a_n$ and 1.

First, we prove these work. Proceed via contradiction; suppose we have some $b \geqslant c$ with $b + c = 2 \cdot 10^{a_n} + 2 \cdot 10^{a_{n-1}} + \dots 2 \cdot 10^{a_1} \pm 1$ such that b, c do not contain the digit 1. It is not hard to see that the leading digit of b must be 2, ie $b = 2 \cdot 10^{a_n} + k$ for some natural k, which means that $c + k = 2 \cdot 10^{a_{n-1}} + 2 \cdot 10^{a_{n-2}} + \dots 2 \cdot 10^{a_1} \pm 1$. Repeating this process recursively, each of $2 \cdot 10^{a_j}$ will be assigned to one of b or c, and adding or subtracting 1 at the end will make one of these two numbers have a 1 appear in their decimal representation; contradiction.

Call a number Ackerman if it is in the form $2 \cdot 10^{a_n} + 2 \cdot 10^{a_{n-1}} + \dots 2 \cdot 10^{a_1} \pm 1$ (or 1), and for the sake of convenience let $2 \cdot 10^k + 2 \cdot 10^{k-1} + \dots 2 \cdot 10^1 + 2 = S_k$.

It suffices to show that all non-Ackerman numbers are non-solitary.

Claim

If all non-Ackerman numbers $\leqslant S_k$ are non-solitary, then all non-Ackerman numbers $\leqslant S_{k+1}$ are non-solitary.

Proof 2.1.6:

If a number x is non-solitary, then so are $10x+2, 10x+3, \dots 10x+11$, as if $x = y+z$, we can rewrite $10x + c$ for some $c \in \{2, 3, \dots 11\}$ as $10y + l + 10z + p$ for some $l, p \in \{0, 2, 3, \dots 9\}$. Consider the set $\{S_{n-1}, S_{n-1} + 1, \dots S_n\}$. It follows that all numbers in the set

$$\{10S_{n-1} + 2, 10S_{n-1} + 3, \dots 10S_n + 11\} \equiv \{S_n, S_n + 1, \dots S_{n+1} + 9\}$$

are non-solitary, set aside the ones in the form $10x + c$ for some Ackerman x and some $c \in \{2, 3, \dots 11\}$. It follows that all of these numbers can be rewritten in the form

$$\sum_{k=1}^{n} 2 \cdot 10^{b_k} + m$$

for some $2 \leqslant b_1 < b_2 < \dots < b_n$, and some $m \in \{-8, -7, \dots 1, 12, 13, \dots 21\}$.

From here we can deduce that the only values of m which yield solitary numbers here is $m = -1, 1, 19, 21$, which all yield Ackerman numbers, finishing the inductive hypothesis. $\square$

As such, when constructing a solitary number $< 10^{2026}$, we effectively have 2026 digit slots and in each one we must place a 0 or 2, and at the end we must choose to add or subtract 1 from this number. Now, the units digit must be 0, as it cannot be 2, which means we have $2^{2025} \cdot 2 = 2^{2026}$ possible solitary numbers, but we must subtract 1 to account for the case where all the digits are 0 and we subtract 1, ie the number of solitary numbers $< 10^{2026}$ is $\boxed{2^{2026} - 1}$, and we are done. $\square$

2.2 USAMO 2026 / Problem 5 Solution

Problem 2.2.1:

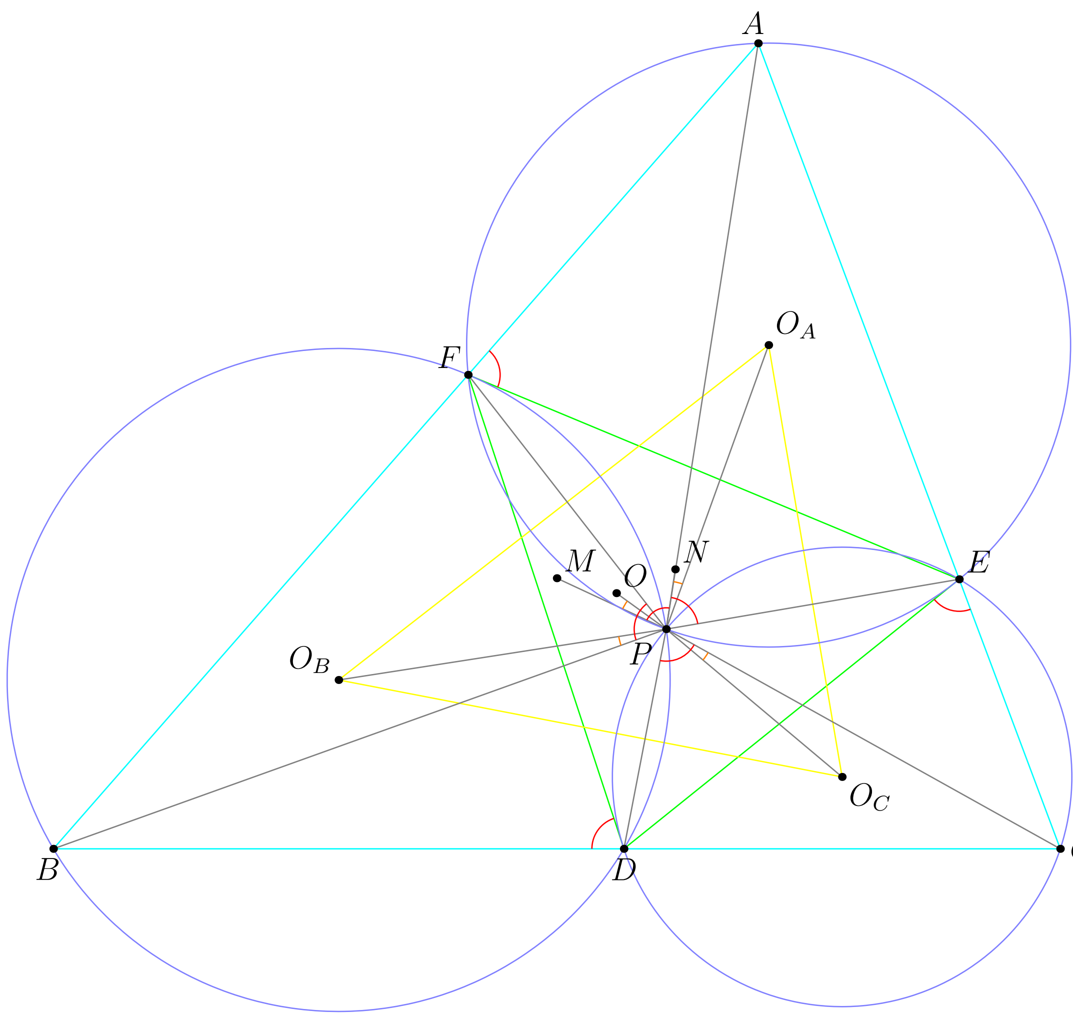
Let ABC be a triangle. Points D, E, and F lie on sides BC, CA, and AB, respectively, such that

$$\angle AFE = \angle BDF = \angle CED.$$

Let O_A, O_B, and O_C be the circumcenters of triangles AFE, BDF, and CED, respectively. Let M, N, and O be the circumcenters of triangles ABC, DEF, and $O_AO_BO_C$, respectively. Prove that $OM = ON$.

★ *Proposed by Serena An*

2.2.1 Solution 1



Let P be the point where the circumcircles of $\triangle AFE, \triangle BDF, \triangle CED$ intersect, which we know exists because it is the Miquel point of the given configuration.

Claim

We have $\triangle ABC \sim \triangle EFD \sim \triangle O_AO_BO_C$.

Proof 2.2.1:

For $\triangle ABC \sim \triangle O_AO_BO_C$, we have that

$$\angle CAB = \angle EAP + \angle PAF = \angle O_CO_AP + \angle PO_AO_B = \angle O_CO_AO_B$$

and

$$\angle BCA = \angle DCP + \angle PCE = \angle O_BO_CP + \angle PO_CO_A = \angle O_BO_CO_A$$

from O_A, O_B, O_C being the centers of circumcircles $\triangle AFE, \triangle BDF, \triangle CED$.
For $\triangle ABC \sim \triangle EFD$, we have from $\angle AFE = \angle BDF = \angle CED$ and D, E, F lying on BC, CA, AB respectively that

$$180^\circ = \angle CED + \angle DEF + \angle FEA = \angle AFE + \angle CAB + \angle FEA \implies \angle DEF = \angle CAB$$

and analogously $\angle FDE = \angle BCA$. $\square$

Claim

P is the center of the spiral similarities s_1, s_2 sending $\triangle ABC$ to $\triangle EFD$ and $\triangle ABC$ to $\triangle O_AO_BO_C$ respectively.

Proof 2.2.2:

For s_1 note by cyclic quadrilaterals $AEPF, CDPE, BFPD$ that

$$\angle AFE = \angle BDF = \angle CED = \angle CPD = \angle BPF = \angle EPA,$$

which we call θ, and $\angle AEP = \angle BFP = \angle CDP$. Hence $\triangle AEP \sim \triangle BFP \sim$

$\triangle CDP$ so s_1 rotates $\triangle ABC$ by θ and scales down by the common ratio

$$\frac{AP}{PE} = \frac{BP}{PF} = \frac{CP}{PD}$$

to precisely yield $\triangle EFD$.

For s_2 note that from $\angle PDC = \angle PEA = \angle PFB$ follows $\angle APO_A = \angle BPO_B = \angle CPO_C$ and $\angle PO_AA = \angle PO_BB = \angle PO_CC$, which means $\triangle APO_A \sim \triangle BPO_B \sim \triangle CPO_C$ and similarly suffices to imply that P is the center of s_2.

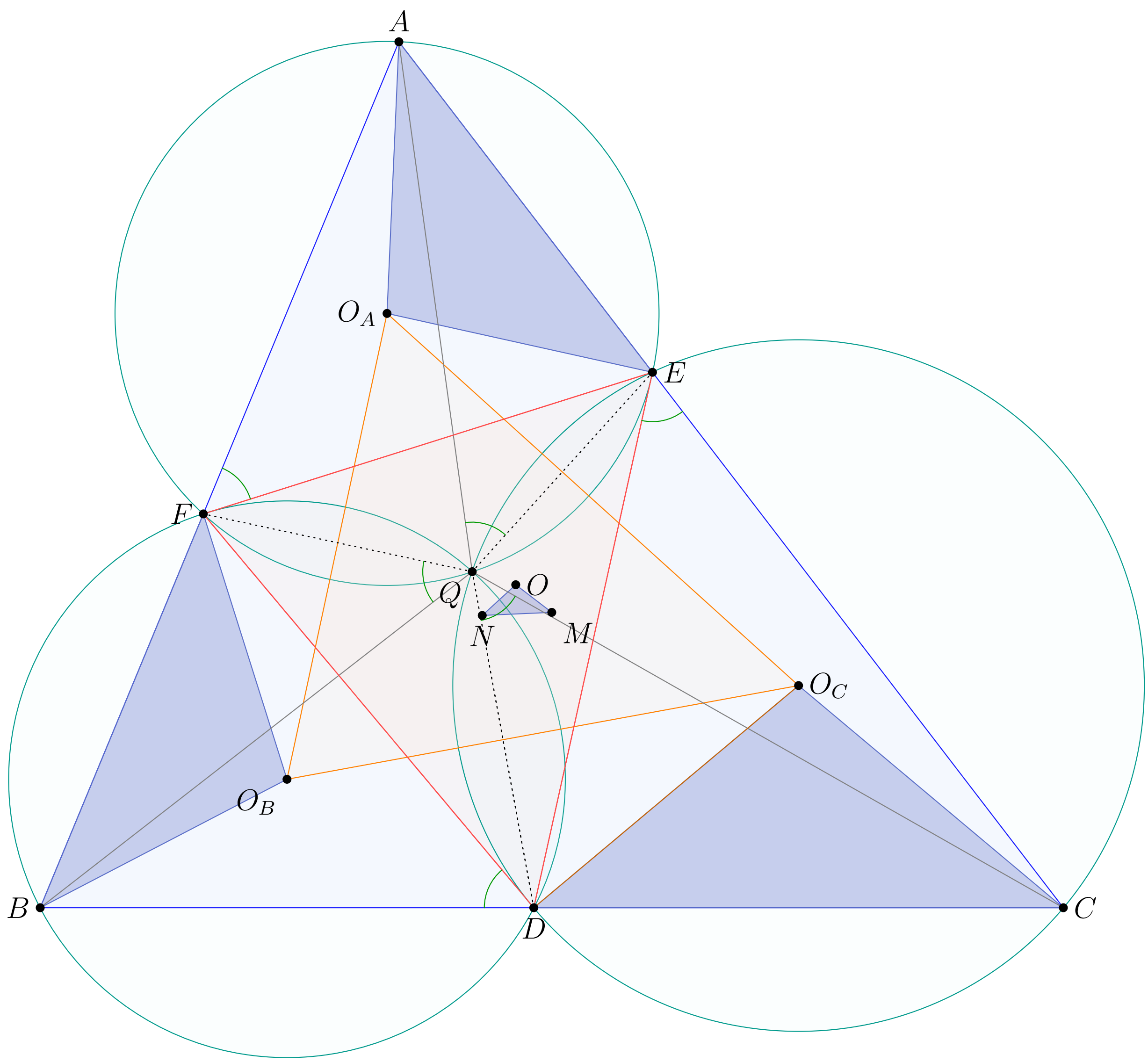
To finish, $s_1(M) = N$ and $s_2(M) = O$. Since $s_1(A) = E$ and $s_2(A) = O_A$ while s_1, s_2 share the same center, $\triangle MON \sim \triangle AO_AE$ from which $OM = ON$ because $O_AA = O_AE$. $\square$

2.2.2 Solution 2

For now, entirely ignore the points M, N, O; we'll deal with them at the end.

By Miquel theorem, the circumcircles of AEF, BFD, and CDE meet at a point Q, the most important point in this solution. Define the angles

$$\alpha = \measuredangle AFE = \measuredangle BDF = \measuredangle CED$$
$$\beta = \measuredangle QEA = \measuredangle QEC = \measuredangle QDC = \measuredangle QDB = \measuredangle QFB = \measuredangle QFA.$$



We will make explicit a spiral similarity from $\triangle ABC$ to $\triangle DEF$ centered at Q. It will be best stated in the following form:

Claim

We have the spiral similarities

$$\triangle QAE \overset{+}{\sim} \triangle QBF \overset{+}{\sim} \triangle QCD.$$

Proof 2.2.3:

Note that $\measuredangle AQE = \measuredangle AFE = \alpha$ (marked in green in the figure), while $\measuredangle QEA = \beta$. $\square$

In fact, now we claim $\triangle O_AO_BO_C$ is similar too.

Claim

The same spiral similarities at Q map

$$\triangle AO_AE \overset{+}{\sim} \triangle BO_BF \overset{+}{\sim} \triangle CO_CD.$$

Proof 2.2.4:

These triangles are isosceles with $\measuredangle AO_AE = 2\alpha$. $\square$

Finally we bring in the points M, N, O used only for extraction. Since our spiral similarities map ABC, DEF, and $O_AO_BO_C$ to each other, and these are the respective circumcenters, we can extend our similarity to

$$\triangle AO_AE \overset{+}{\sim} \triangle BO_BF \overset{+}{\sim} \triangle CO_CD \overset{+}{\sim} \triangle MON.$$

In particular, $OM = ON$.

Remark

The choice of M, N, O as the *circumcenters* (as opposed to any other center) of the three triangles ABC, DEF, $O_AO_BO_C$ is completely superficial and has no effect on the proof. That is, one could have replaced those with any triangle center and the proof would carry over verbatim.

Remark

Actually $\measuredangle QAE = \measuredangle QFE = \measuredangle QFA - \alpha = \beta - \alpha$, so by symmetry we actually also know

$$\measuredangle QAE = \measuredangle QBF = \measuredangle QCD = \beta - \alpha.$$

So the point Q is the so-called second Brocard point of the triangle.

2.2.3 Solution 3

We skip up to X, the Miquel Point of ABC with respect to D, E, F, being the center of the spiral similarity from ABC to DEF. Use complex bashing with X as the origin, and let $A = a, B = b, C = c$. Then, since $A \to E, B \to F, C \to D$, we have $E = ka, F = kb, D = kc$ for some fixed complex number k.

Recall

$$O(x, y, z) = \frac{\det \begin{bmatrix} 1 & x & x\overline{x} \\ 1 & y & y\overline{y} \\ 1 & z & z\overline{z} \end{bmatrix}}{\det \begin{bmatrix} 1 & x & \overline{x} \\ 1 & y & \overline{y} \\ 1 & z & \overline{z} \end{bmatrix}}$$

Now, $O_a = O(X, A, E) = O(0, a, ka)$

$$= \frac{\det \begin{bmatrix} 1 & a & a\overline{a} \\ 1 & ka & ka\overline{k}\overline{a} \\ 1 & 0 & 0 \end{bmatrix}}{\det \begin{bmatrix} 1 & a & \overline{a} \\ 1 & ka & \overline{k}\overline{a} \\ 1 & 0 & 0 \end{bmatrix}}$$

Each matrix has only one non-zero minor. The top determinant equals $a^2 k\overline{a}(\overline{k} - 1)$, and the bottom equals $a\overline{a}(\overline{k} - k)$. So,

$$O_a = a\frac{k\overline{k} - k}{\overline{k} - k} \quad \text{etc.}$$

Let $M = m$. Then, $N = km$ by the spiral similarity. Finally, $O = O(O_a, O_b, O_c) = \frac{k\overline{k} - k}{\overline{k} - k}O(a, b, c) = \frac{k\overline{k} - k}{\overline{k} - k}m$. Now, $O - M = m(\frac{k\overline{k} - k}{\overline{k} - k} - 1)$, and $O - N = m\frac{k\overline{k} - k}{\overline{k} - k} - k$. To show these have the same magnitude, it suffices to show that $|\frac{k\overline{k} - k}{\overline{k} - k} - 1| = |\frac{k\overline{k} - k}{\overline{k} - k} - k|$. Equivalently, $|k\overline{k} - k - \overline{k} + k| = |\overline{k}k - k - k\overline{k} + k^2|$ $|k\overline{k} - \overline{k}| = |k^2 - k|$ Squaring both sides, $(k\overline{k} - \overline{k})(k\overline{k} - k) = k(k - 1)\overline{k}(\overline{k} - 1)$ which is true. $\square$

2.3 USAMO 2026 / Problem 6 Solution

Problem 2.3.1:

Let a and b be positive integers such that $\varphi(ab+1)$ divides a^2+b^2+1. Prove that a and b are Fibonacci numbers.

Note: Euler's totient function $\varphi(n)$ is the number of positive integers less than or equal to n that are relatively prime to n.

★ *Proposed by Raymond Feng and Eric Shen*

2.3.1 Solution 1

We start with the following.

Claim
The number $ab+1$ must be a prime power.

Proof 2.3.1:

Note to start that:

- Since Euler phi only gives even values, $2 \mid a^2+b^2+1$.
- Hence, exactly one of a and b is odd.
- Thus, $a^2+b^2+1 \equiv 2 \pmod 4$.

So $\nu_2(\varphi(ab+1)) = 1$, and hence $ab+1$ must have at most one prime factor. □

Henceforth, let $ab+1 = p^e$. The $e=1$ case is easy, and does not even rely on $ab+1$ being prime:

Claim
The equation $ab+1 \mid a^2+b^2+1$ is equivalent to $\{a,b\} = \{F_{2k-1}, F_{2k+1}\}$ for $k \geq 0$. Here the Fibonacci sequence F_i is indexed by $F_{-1} = F_1 = F_2 = 1$ and $F_0 = 0$.

Proof 2.3.2:

Classical Vieta jumping argument. $\square$

We now consider $e \geq 2$. In that case, we have

$$\varphi(p^e) = p^{e-1} \cdot (p-1) \mid a^2 + b^2 + 1$$
$$p^e - 1 = ab.$$

This implies that

$$0 \equiv a^2 + \left(\frac{p^e - 1}{a}\right)^2 + 1 \equiv \frac{a^4 + a^2 + 1}{a^2} = \frac{(a^2 + a + 1)(a^2 - a + 1)}{a^2} \pmod{p^{e-1}}.$$

In particular, this implies $p = 3$ or $p \equiv 1 \pmod 3$.

- If $p \equiv 1 \pmod 3$, then we get $3 \mid p - 1 \mid a^2 + b^2 + 1$, which forces a and b to be nonzero modulo 3. However, then $p^e - 1 = ab \not\equiv 0 \pmod 3$, which is impossible.
- Now suppose $p = 3$. Since $x^2 \pm x + 1$ is never divisible by 9, this forces $e \leqslant 2$. Hence, we only have to consider the case $ab = 8$. We find there is one additional solution here only: $(1, 8)$, and both 1 and 8 are Fibonacci numbers.

2.3.2 Solution 2

Corollary 2.3.1:

n is a Fibonacci number if and only if at least one of $5n^2-4$ and $5n^2+4$ is a perfect square.

All of $1, 2, 3$ are Fibonacci numbers hence assume that $\max\{a, b\} \geqslant 4$.

Claim

$ab+1$ is a prime power.

Proof 2.3.3:

Since $\varphi(ab+1)$ is even, $2 \mid a^2+b^2+1$ thus, exactly one of a, b is even. If $ab+1 = p_1^{\alpha_1} \dots p_\ell^{\alpha_\ell}$ where $p_i > 2$, then by modulo 4,

$$1 \geqslant \nu_2(a^2+b^2+1) \geqslant \nu_2(\varphi(ab+1)) = \nu_2(p_1-1) + \cdots + \nu_2(p_\ell-1) \geqslant \ell$$

so the result follows. $\square$

Let $ab+1 = p^k$ for a prime $p > 2$.

★ **Case 1**: $k \geqslant 2$

We have $p^{k-1}(p-1) \mid a^2+b^2+1$. Since $p \mid ab+1$ and $p \mid a^2+b^2+1$, we get $p \mid a^2-ab+b^2$ which implies $p \mid (2a-b)^2+3b^2$. If $p \neq 3$, then $\left(\frac{p}{3}\right) = \left(\frac{-3}{p}\right) = 1$ however, $3 \mid p-1$ is impossible because otherwise $3 \mid ab$ and $3 \mid a^2+b^2+1$. Hence $p = 3$. We observe that $3^{k-1} \mid a^4+a^2b^2+a^2$ so $3^{k-1} \mid a^4+a^2+1 = \frac{a^6-1}{a^2-1}$. Since $3 \nmid a$, by LTE

$$k-1 \leqslant \nu_2(a^6-1) - \nu_2(a^2-1) = 1$$

thus, $k=2$. This yields $\{a, b\} = \{1, 8\}$ which are Fibonacci numbers. $\square$

★ **Case 2**: $k = 1$

Let's start by presenting a claim.

Claim

Vieta Jumping Claim: If $xy \mid x^2 + y^2 + 1$, then $\dfrac{x^2+y^2+1}{xy} = 3$.

Proof 2.3.4:

Assume that $x^2 - kxy + y^2 + 1 = 0$ has a positive integer solution for $k \neq 3$. Let (x, y) be the solution with minimal sum. WLOG suppose that $x \geqslant y$.

$$x^2 - (ky)x + y^2 + 1 = 0$$

Let x' be the other root. Explicitly $x' \geqslant x$. We have $x + x' = ky$ and $xx' = y^2 + 1$. Notice that $x' \neq x$ because $y^2 - x^2 \neq 1$. Thus, $x' \geqslant x + 1$. This indicates $y^2 + 1 = xx' \geqslant x^2 + x \geqslant y^2 + x$ so $x = 1 = y$ which requires $k = 3$. □

Hence $a^2 + b^2 + 1 = 3ab$ which is $b^2 - (3a)b + a^2 + 1 = 0$. Consequently, $\Delta = 9a^2 - 4a^2 - 4 = 5a^2 - 4$ which must be a perfect square and similarily $5b^2 - 4$ is a perfect square. By the corollary, both a, b must be Fibonacci numbers as desired. □

2.3.3 Solution 3

Let $F_1 = 1, F_2 = 1, F_3 = 2, \ldots$ denote the Fibonacci numbers. The proof will proceed in several steps.

Claim

We have $ab + 1 = p^n$ for some prime p and positive integer n.

Proof 2.3.5:

Suppose to the contrary that $ab+1$ has k distinct prime divisors $p_1, p_2, \ldots, p_k$, with $k \geqslant 2$. Then $2^k \mid (p_1 - 1)(p_2 - 1) \cdots (p_k - 1) \mid \varphi(ab+1) \mid a^2 + b^2 + 1$, a contradiction since $a^2 + b^2 + 1 \in \{1, 2, 3\} \pmod 4$. Thus $k = 1$ as intended. $\square$

Examine the case $p = 2$. Since $4 \mid \varphi(2^n)$ for $n \geqslant 3$, we must have $n \in \{1, 2\}$. Checking both cases produces $(a, b) = (1, 1), (1, 3)$ and permutations, which both satisfy the desired property. Henceforth we assume p is odd.

Claim

Either $p = 3$ or $n = 1$.

Proof 2.3.6:

Suppose $n > 1$, so that $p \mid p^{n-1}(p-1) = \varphi(ab+1) \mid a^2 + b^2 + 1$. Assume $p > 3$. Since $ab + 1 \equiv 0 \pmod p$ and clearly $p \nmid a, b$, we can express $b \equiv -a^{-1} \pmod p$. This gives

$$a^2 + a^{-2} + 1 \equiv 0 \pmod p \implies (a + a^{-1} + 1)(a + a^{-1} - 1) \equiv 0 \pmod p,$$

implying $a^2 \pm a + 1 \equiv 0 \pmod p$. Rewrite this as $(2a \pm 1)^2 \equiv -3 \pmod p$, so that $\left(\frac{-3}{p}\right) = 1$. We remark that $p \equiv 3 \pmod 4$, since otherwise $4 \mid p - 1 \mid \varphi(p^n)$. This implies

$$1 = \left(\frac{-3}{p}\right) = \left(\frac{-1}{p}\right)\left(\frac{3}{p}\right) = -\left(\frac{3}{p}\right) = -(-1)^{\frac{p-1}{2}}\left(\frac{p}{3}\right) = \left(\frac{p}{3}\right)$$

by quadratic reciprocity, so $p \equiv 1 \pmod 3$. But then $3 \mid p-1 \mid a^2+b^2+1$, which gives $3 \nmid a, b$. This is a contradiction since $ab = p^n - 1 \equiv 0 \pmod 3$. Thus $p = 3$ as desired. $\square$

Observe that if $p = 3$, taking $n = 1, 2$ yields the solutions $(1, 2)$ and $(1, 8)$, both pairs of Fibonacci numbers. Otherwise $n \geqslant 3$, from which we deduce $9 \mid \varphi(3^n)$ and hence $a^2+b^2 \equiv -1 \pmod 9$. But $ab \equiv -1 \pmod 9$, and checking all 3 pairs produces $a^2 + b^2 \equiv 2 \pmod 9$, a contradiction. We now restrict our attention to $n = 1$, i.e. $ab = p - 1$.

Claim

If $ab + 1 = p$ for a prime $p > 2$, then $(a, b) = (F_{k+2}, F_k)$ for some odd integer $k \geqslant 1$.

Proof 2.3.7:

Note that $\varphi(ab + 1) = p - 1 = ab$, so $ab \mid a^2 + b^2 + 1$. This is Vieta jumping; write $a^2+b^2+1 = qab$ for some positive integer q, so that $a^2-qba+(b^2+1) = 0$. Suppose (a, b) satisfies this equation with $a \geqslant b > 0$, and assume $a + b$ is minimal. We are given that

$$(qb - a, b) = \left(\frac{b^2 + 1}{a}, b\right)$$

is another integer solution to the equation, but

$$\frac{b^2 + 1}{a} \leqslant \frac{(a - 1)^2 + 1}{a} < a$$

as long as $a > 1$, which readily yields $(a, b) = 1$ and $q = 3$. By an easy induction we can prove that $F_k^2 + (-1)^{k+1} = F_{k+2}F_{k-2}$ for $k \geqslant 3$, so by repeatedly iterating $(a, b) \to (a, \dfrac{a^2 + 1}{b})$ for $a \geqslant b$ we obtain the chain

$$(1, 1) \to (2, 1) \to (5, 2) \to (13, 5) \to \cdots \to (F_{k+2}, F_k) \to \cdots .$$

This establishes the result. $\square$

Since all preconsidered cases result in a pair of Fibonacci numbers, we conclude. $\square$